Hao Zhen

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Summary

Machine Learning Engineer / Researcher specializing in **LLM and VLM fine-tuning** for real-world, safety-critical multimodal analytics. Hands-on experience with **supervised fine tuning**, **reinforcement learning**, **prompting + evaluation**, and building end-to-end pipelines across text, tabular, and vision data. Published work on **domain-adapted LLMs**, **video VLM fine-tuning**, and **interpretable ML** for large-scale transportation systems.

Education

- **University of Georgia** — Athens, GA
  Ph.D., Civil and Environmental Engineering (Advisor: Dr. Jidong J. Yang) — *Aug 2021 – Aug 2025*
- **North China Electric Power University** — Beijing, China
  M.S., Technological Economics and Management — *Sep 2018 – Jun 2021*
- **North China Electric Power University** — Beijing, China
  B.Eng., Electrical Engineering — *Sep 2014 – Jun 2018*

Experience

- **Postdoctoral Researcher (Multimodal Traffic Foundation Models)** — Oct 2025 – Present
  University of Georgia — *Athens, GA*
  - Lead research on **LLM/VLM-based safety understanding** from crash narratives, multi-view imagery, and structured records.
  - Build training/evaluation workflows for **domain adaptation** and **interpretable** foundation-model outputs.
- **Graduate Research Assistant (LLM/VLM + Applied ML)** — Aug 2021 – Aug 2025
  University of Georgia — *Athens, GA*
  - Fine-tuned and evaluated **domain LLMs** for contextual crash analysis and outcome inference; emphasized reliability and explanation.
  - Developed multimodal systems spanning **text + tabular + vision**, including frontier VLMs/LLMs for structured reasoning tasks.
  - Applied representation learning and graph-based modeling on large sensing networks (1000+ sensors; 10K+ road segments) for downstream safety/operations tasks.
- **Graduate Teaching Assistant** — 2024 – 2025
  University of Georgia — *Athens, GA*
  - TA for Advanced Deep Learning (ENGR 8140) and Statistical/Machine Learning (ENGR 8130); created hands-on PyTorch training labs.

Selected Projects (LLM/VLM Focus)

- **CrashSage (Domain LLM + Interpretability)** [paper]: SFT a **LLaMA3-8B** model for contextual crash analysis; achieved **+5.12%** accuracy improvement and added **gradient-based word-level explanations**.
- **MP-PVIR (Fine-tuned VLM for Multi-view Incident Reasoning)** [paper]: Fine-tuned **Qwen-2.5-VL** for incident reasoning and phase segmentation; achieved **64.70% VQA accuracy** and **0.4881 mIoU** on WTS.
- **Crash Diagram Generation (VLM Benchmarking + Structured Output)** [paper]: Designed evaluation for diagram generation from reports on multi-lane roundabouts; benchmarked GPT-4o, Gemini, and Janus-4o.
- **Tab-Text (Tabular-to-Text for Safety Modeling)** [paper]: Bridged crash tables with narratives by fine-tuning **ELECTRA**; improved inference for rare severe outcomes (fatal/serious injury).
- **Feature Group Tabular Transformer (Structured Tokens for Causal Analysis)** [paper]: Organized heterogeneous features into semantic groups as tokens to improve interpretability and uncover collision patterns.

Selected Publications (Google Scholar)

1. **Zhen, H.**, Yang, J.J. (2025). *CrashSage: A Large Language Model-Centered Framework for Contextual and Interpretable Traffic Crash Analysis*. Artificial Intelligence for Transportation.
   (Domain LLM, SFT, interpretability)
2. **Zhen, H.**, Yang, J.J. (2025). *Tab-Text: Bridging Tabular Data and Natural Language for Enhanced Traffic Safety Analysis*. Expert Systems with Applications.
   (Tabular–Text multimodal learning, fine-tuned language models)
3. **Zhen, H.**, Shi, Y., Huang, Y., Yang, J.J., Liu, N. (2024). *Leveraging Large Language Models with Chain-of-Thought and Prompt Engineering for Traffic Crash Severity Analysis*. Computers.
   (Prompt engineering, CoT reasoning, LLM evaluation)
4. **Zhen, H.**, Yang, J.J. (Accepted, TRB 2026). *Multi-view Phase-aware Pedestrian–Vehicle Incident Reasoning with Vision-Language Models*.
   (VLM fine-tuning, Qwen-2.5-VL, multimodal reasoning)
5. Lu, X., **Zhen, H.**, Yang, J.J. (Accepted, TRB 2026). *Automating Crash Diagram Generation Using Vision-Language Models*. Transportation Research Board (TRB)
   (Structured output generation, VLM benchmarking)
6. **Zhen, H.**, Yang, J.J. (2024). *Analyzing the Importance of Network Topology in AADT Estimation using Graph Neural Networks*. Transportation.
   (Graph representation learning)
7. Lares, O., **Zhen, H.**, Yang, J.J. (2025). *Feature Group Tabular Transformer: A Novel Approach to Traffic Crash Modeling and Causality Analysis*. Applied Computing and Intelligence.
   (Structured tokens, interpretability, causality-oriented modeling)
8. **Zhen, H.**, Yang, J.J. (Ongoing). *Beyond Correlation: Enhancing Causal Reasoning in Traffic Crash Analysis using Group Relative Policy Optimization (GRPO) for Large Language Models*.
   (LLM alignment, preference optimization)
9. **Zhen, H.**, Yang, J.J. (Ongoing). *Cognitive Multimodal Models for Human–Autonomy Collaboration in Connected Vehicles*.
   (Multimodal reasoning, human-aligned AI)
10. Yang, Y., **Zhen, H.**, Yang, J.J. (Ongoing). *Reinforcement Learning for Multimodal Reasoning: Human-Aligned Vision-Language Models for Intersection Safety*.
    (RL + VLM alignment)

Additional publications (17+ total) in graph learning, time-series forecasting, optimization, and energy systems are available on Google Scholar.

Technical Skills

- **LLM/VLM:** supervised fine-tuning (SFT), PEFT, prompt engineering, agent training, evaluation;
- **ML/Engineering:** PyTorch, TensorFlow, scikit-learn; dataset curation, experiment tracking
- **Tooling:** Transformers, Hugging Face ecosystem, OpenAI API, TRL, vLLM, LangChain; Docker, Git, Weights & Biases; AWS (EC2/S3/SageMaker)
- **Core languages:** Python, Shell; (also R, MATLAB)

Leadership & Recognition

- Certificate of Appreciation, USDOT Region 4 University Transportation Center (CR2C2), 2025
- APHA Injury Control and Emergency Health Services Student Scholarship, 2025; WTS Helene M. Overly Memorial Scholarship, 2022
- Program Committee: PAKDD 2026; Reviewer: IEEE T-VT, Neural Computing & Applications